The role of global magnetic coupling in exoplanet atmosphere: Insights from star-planet magnetic interactions

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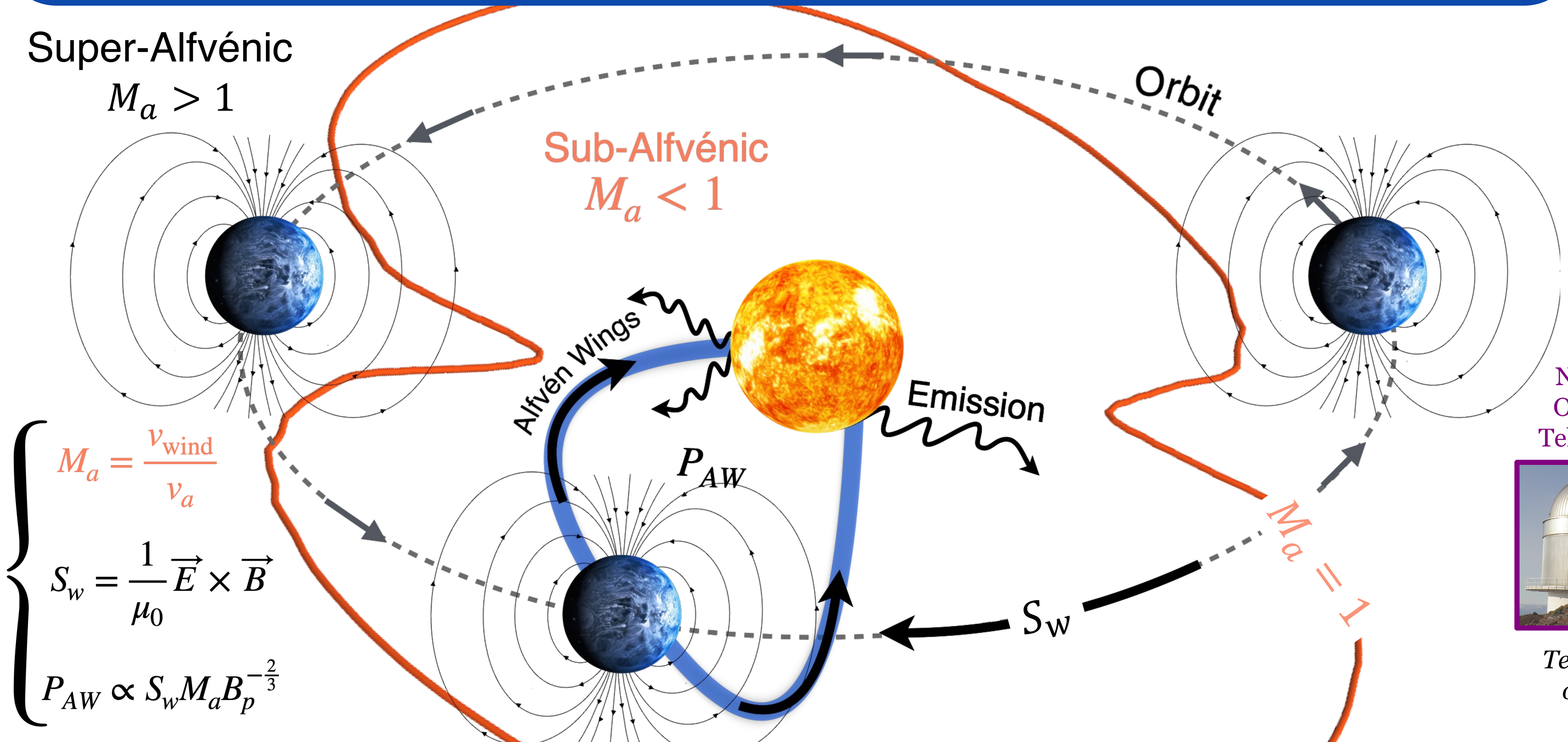
Context

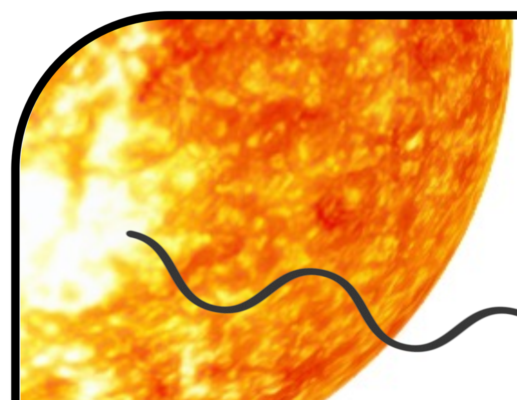
Exploring the diversity of planetary atmospheres calls for sophisticated models that combine photochemistry, atmospheric dynamics, and interactions with the host star. To investigate magnetic coupling in exoplanetary atmospheres, we need tighter constraints on their **magnetic fields**. Yet, not a single exoplanetary magnetic field has been fully characterised at the moment.

➢ We model stellar winds of HD 189733 system to investigate conditions for **Star-Planet Magnetic Interactions** (SPMI) and their potential as an indirect probe of **exoplanetary magnetic fields**.

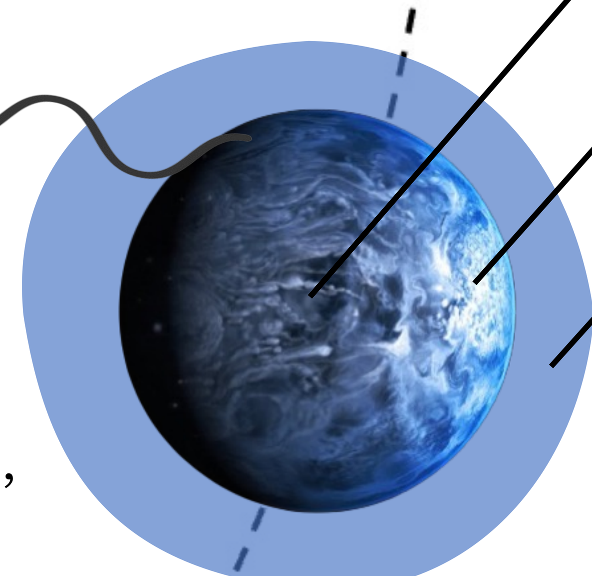


$$\begin{cases} M_a = \dfrac{v_{\text{wind}}}{v_a} \\ S_w = \dfrac{1}{\mu_0}\vec{E}\times\vec{B} \\ P_{AW} \propto S_w M_a B_p^{-\frac{2}{3}} \end{cases}$$

Zoom on the HD189733 system



Modulation of stellar activity, linked to the orbital period[3,4]

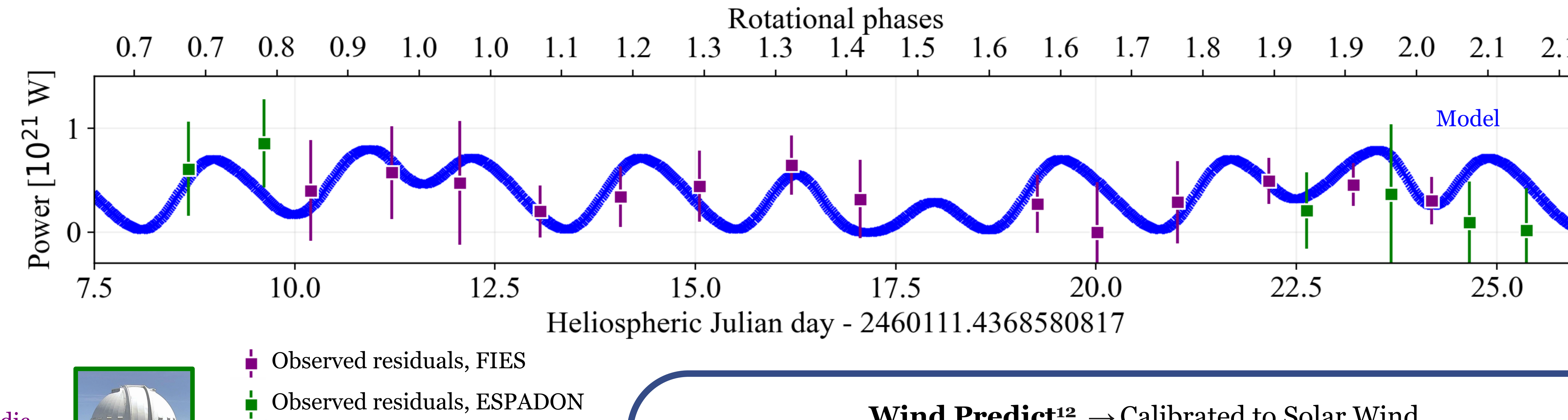


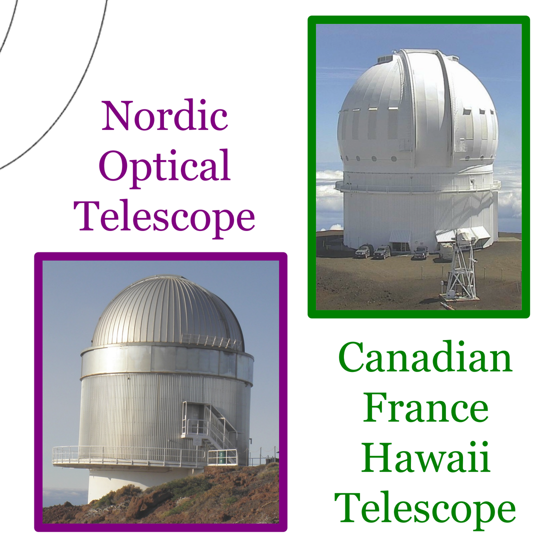
- Hot Jupiter, discovered in 2005[1]
- Transiting, Porb ~ 2.2 days[2]
- Not in the habitable zone

A rich atmosphere, undergoing atmospheric escape[5]

➢ Water[6], Methane[7], Sodium[8], Potassium[9], CO_2[10], CO[10] and H_2S[10] discovered

- ✓ Observation of the stellar magnetic field at the surface of the star in Zeeman-Doppler imaging to **constrain the stellar magnetic field**.
- ✓ Spectro-polarimetric **observational campaign** led in 2023 in K-band photometry by Dr. Strugarek, Moutou, Fares and Yang.
- ✓ Need to **model the 3D environment** of the star : use of PLUTO[11] to resolve MHD equations in order to characterise stellar winds.



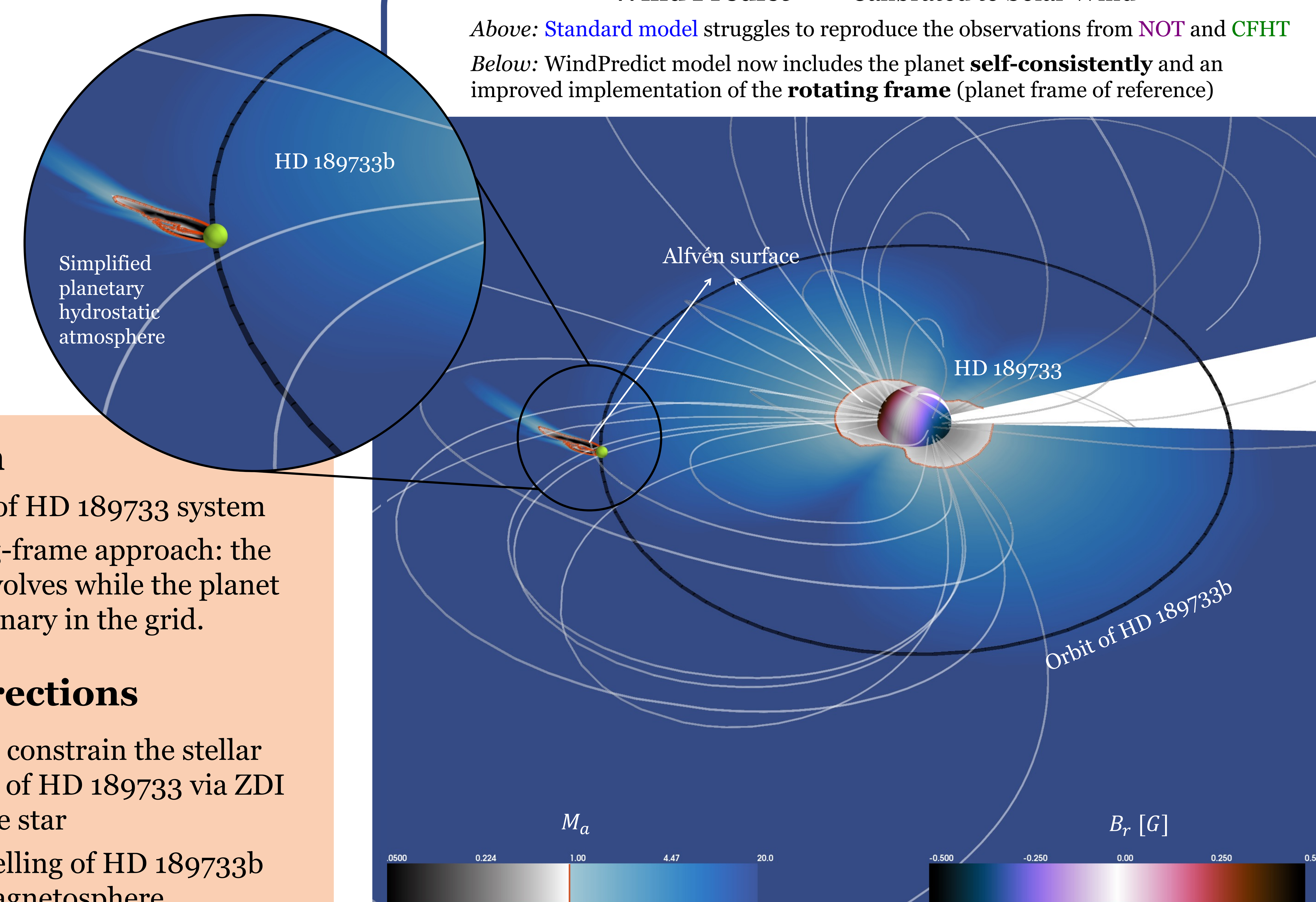


Telescopes used for the observations of HD 189733 CaII

Wind Predict[12] → Calibrated to Solar Wind

Above: Standard model struggles to reproduce the observations from NOT and CFHT

Below: WindPredict model now includes the planet **self-consistently** and an improved implementation of the **rotating frame** (planet frame of reference)



Conclusion

- ✓ Modelisation of HD 189733 system
- ✓ Novel rotating-frame approach: the astrosphere evolves while the planet remains stationary in the grid.

Futures directions

- ❑ Need to better constrain the stellar magnetic field of HD 189733 via ZDI mapping of the star
- ❑ Realistic modelling of HD 189733b including a magnetosphere

Abstract

References:

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